Step 1: Cylindrical Query Initialization
Initialize a set of 3D cylindrical queries across the BEV.
Each query encodes an object hypothesis.
ImV
ImV
BEV
Step 2: Feature Sampling & Aggregation
Project each 3D cylindrical query into all camera views.
Aggregate multi-view image features .
ImV
ImV
BeV
Step 3: Cylinder Position & Shape Refinement
Refine cylindrical query using the aggregated multi-view
features, ensuring improved BEV and ImV alignment.
ImV
ImV
BEV
MVCyDe:
3D-BeV & 2D Image-view Detection & Tracking
C1
C7
C3
BEV
ImV
ImV
ImV
BEV: Bird's-Eye-View
ImV: Image-View
Projection Direction
Absolute Ground Point
3D Cylindrical Query
Projected bounding box from
Cylindrical Query